

## **Development Of A Microelectrode Array With Semitransparent Carbon Nanotube Electrodes for In-Vitro Applications And Optogenetics**

Manuel Martina<sup>1</sup>, Kathrin Besca<sup>1</sup>, P. Kshirsagar<sup>1</sup>, Sandra Buckenmaier<sup>1</sup>, Kerstin Schneider<sup>2</sup>, M. Fleischer<sup>2</sup>, D.P. Kern<sup>2</sup>, Udo Kraushaar<sup>1</sup>, A. Stett<sup>1</sup>, C. J. Burkhardt<sup>1</sup>

<sup>1</sup>NMI Natural and Medical Sciences Institute at the University Tübingen, 72770 Reutlingen (Germany)

<sup>2</sup>Institute for Applied Physics and Center LISA+, Eberhard Karls University Tübingen, 72074 Tübingen (Germany)

manuel.martina@nmi.de

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### **Motivation**

Commonly, microelectrode materials are selected in consideration of their electrical interface properties, chemical stability and biocompatibility. With these points in mind, there is a small set of standard microelectrode materials available, namely gold, platinum and titanium nitride. Although there are some uncommon materials, e.g. PEDOT:PSS, graphite and diamond, none of these materials offer a reasonable optical transparency.

However, in several applications electrode transparency is an integral necessity. For example, microfluidic channels and fluid in-/outlets commonly build up on MEA substrates and thus, optical imaging from the top with high resolution is impractical. Here, transparent electrodes allow to observe cell cultures and tissue from the bottom.

Furthermore, in the field of optogenetics transparent electrodes enable light stimulation of the cells in the very contact with the electrode through the substrate [1], as demonstrated by the CLEAR device [2]. There, an all transparent microelectrode array was fabricated out of graphene and successfully utilized.

Here, we present a transparent microelectrode array for in-vitro applications, which is based on a thin layer of carbon nanotubes as electrode interface.

### **Material and Methods**

A MEA substrate was fabricated utilizing indium-tin-oxide conducting paths and silicon oxide insulator with established cleanroom processes. After opening the electrodes and connection pads by reactive ion etching, a thin titanium nitride layer as a diffusion barrier followed by an ultra-thin catalyst layer of nickel is deposited. After lift-off, thermal CNT deposition is performed in a mixed ammonia-acetylene atmosphere [3]. Substrate transparency after different process steps is measured using an UV-vis spectrometer and impedance measurements in PBS are performed using a MCS MEA2100 system and a platinum reference electrode.

For cell culture experiments, ventricular myocytes were isolated from 14 days old embryonic chick hearts as previously described [4]. Field action potentials (fAP) of spontaneously beating cells were recorded after five days in culture at 37°C, 5% CO<sub>2</sub>.

### **Results and Discussion**

Following the described process, MEAs with transparent microelectrodes were fabricated. It was observed, that the thickness of the catalyst layer and the CNT height are the critical factors which determine the morphology and thus the optical transparency of the electrode

layer. After process optimization, the transparency was measured to be around 40% in the visible range, see Figure 1. The impedance of these MEAs (electrode diameter 30  $\mu\text{m}$ ) was measured in the range of 100 k $\Omega$ .

Cultivation of ventricular myocytes on these MEAs was successful. fAPs could be measured with a good signal-to-noise ratio. The basic noise level of the electrodes was around  $\pm 10 \mu\text{V}$  while the signals could be recorded up to an amplitude of 2,5 mV.

The advantage of these MEAs in comparison to the standard MEAs is the transparency of the electrodes which allows visualizing the plated cells directly on the respective electrodes, see Figure 2.

## Conclusions and Outlook

We developed a process to fabricate transparent microelectrodes based on carbon nanotubes. These MEAs were successfully used to measure field action potentials from ventricular myocytes and allowed optical cell imaging from the bottom through the electrode.

We will further improve the transparency and electrical properties of these MEAs and investigate other carbon materials, e.g. graphene, for their feasibility as electrode material.

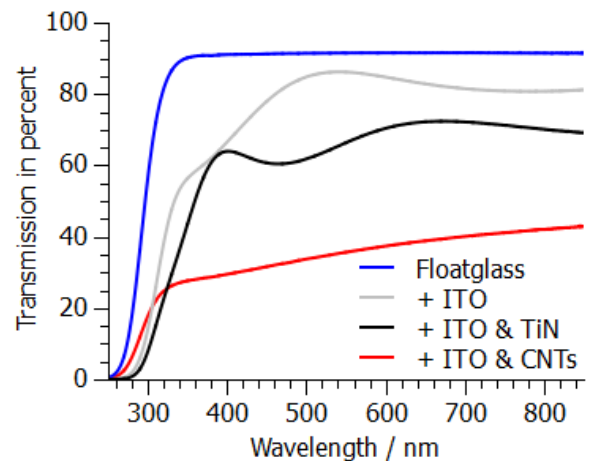


Figure 1: Measured transparency of the substrate after different process steps.

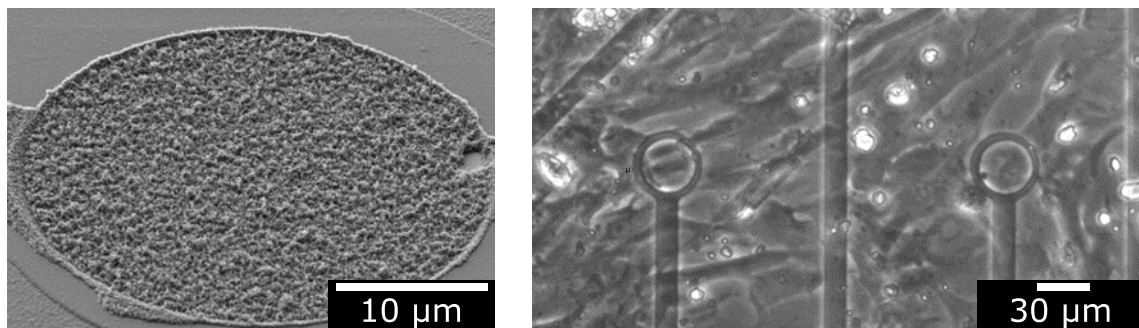


Figure 2: SEM image of a CNT microelectrode (left) and optical image of cardio cells through the electrode (right).

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## References:

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